Formula 1 Network Analysis: Structural Properties, Community Detection, and Robustness in Driver–Constructor Networks

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Project Overview

- Analyze Formula 1 racing data (1950–2020) using network science
- Model driver–constructor relationships as a complex network
- Dataset: Kaggle F1 World Championship (26,759 results, 861 drivers, 211 constructors)

Five Deliverables

1. Bipartite Network Construction
2. Network Projection & Structural Analysis
3. Centrality Analysis & Hub Identification
4. Community Detection
5. Network Robustness Simulation

Deliverable 1: Bipartite Network Construction

- Constructed driver–constructor bipartite graph from `results.csv`
- 861 unique drivers mapped to 211 unique constructors
- Each edge represents a driver's participation with a constructor

Metric	**Value**
TotalNodes	1,072
Driver Nodes	861
Constructor Nodes	211
Total Edges	2,150

Table 1: Bipartite network statistics.

Deliverable 2: Network Projection

- Projected bipartite graph into driver–driver co-occurrence network
- Edges represent shared constructors between driver pairs

Metric	**Value**
Nodes (Drivers)	861
Edges (Shared teams)	1,567
Average Degree	3.64
Density	0.0042

Table 2: Driver network statistics.

Degree Distribution

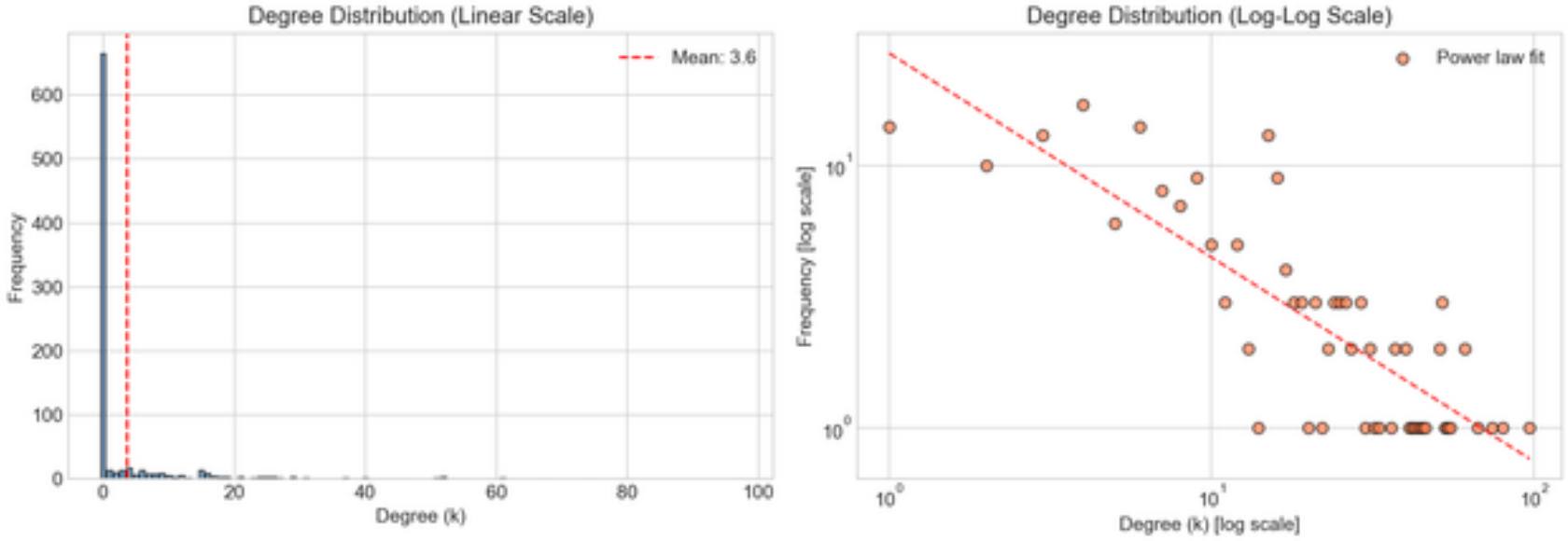


Figure 1: Degree distribution (linear and log-log scale) with power-law fit ($\gamma = 0.78$).

Real Network vs. Random Network

Metric	**Real Network**	**Random Network**
Clustering Coeff. (C)	0.6753	0.0036
Avg Path Length (L)	2.5227	5.3265
$\ln(N)$	6.7581	–

Table 3: Small-world metrics comparison.

- $L = 2.52 < \ln(N) = 6.76 \Rightarrow$ Small-worldpropertyverified
- $C = 0.68 \gg C_{random} = 0.004 \Rightarrow$ Highlocalclustering

Real vs. Random: Visualization

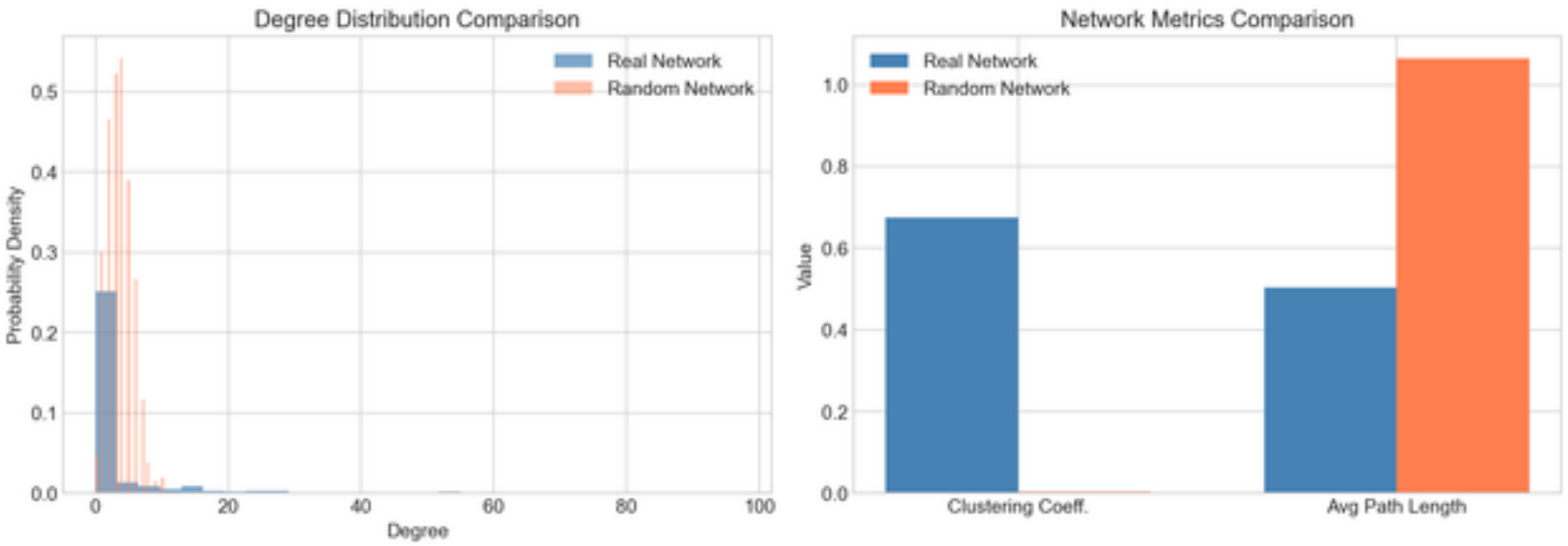


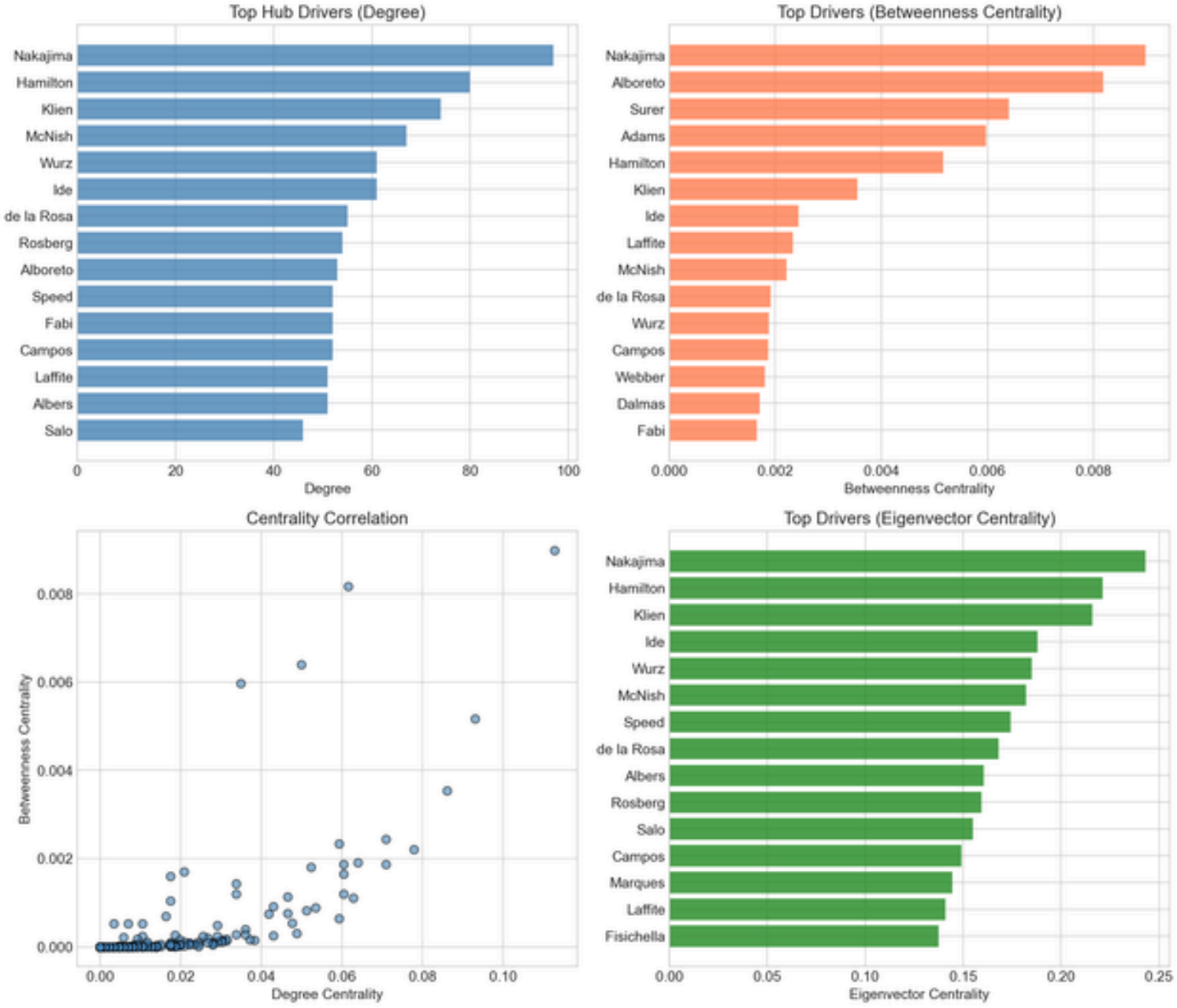
Figure 2: Comparison of real F1 driver network with equivalent random network.

Top 15 Hub Drivers

ID	Name	Degree	BC	EC
6	Nakajima	97	0.0090	0.2432
1	Hamilton	80	0.0052	0.2212
32	Klien	74	0.0035	0.2159
66	McNish	67	0.0022	0.1822
25	Wurz	61	0.0019	0.1850
34	Ide	61	0.0024	0.1878
37	delaRosa	55	0.0019	0.1682
3	Rosberg	54	0.0011	0.1595

Table 4: Top hub drivers. BC = Betweenness, EC = Eigenvector Centrality.

Centrality Analysis: Visualization



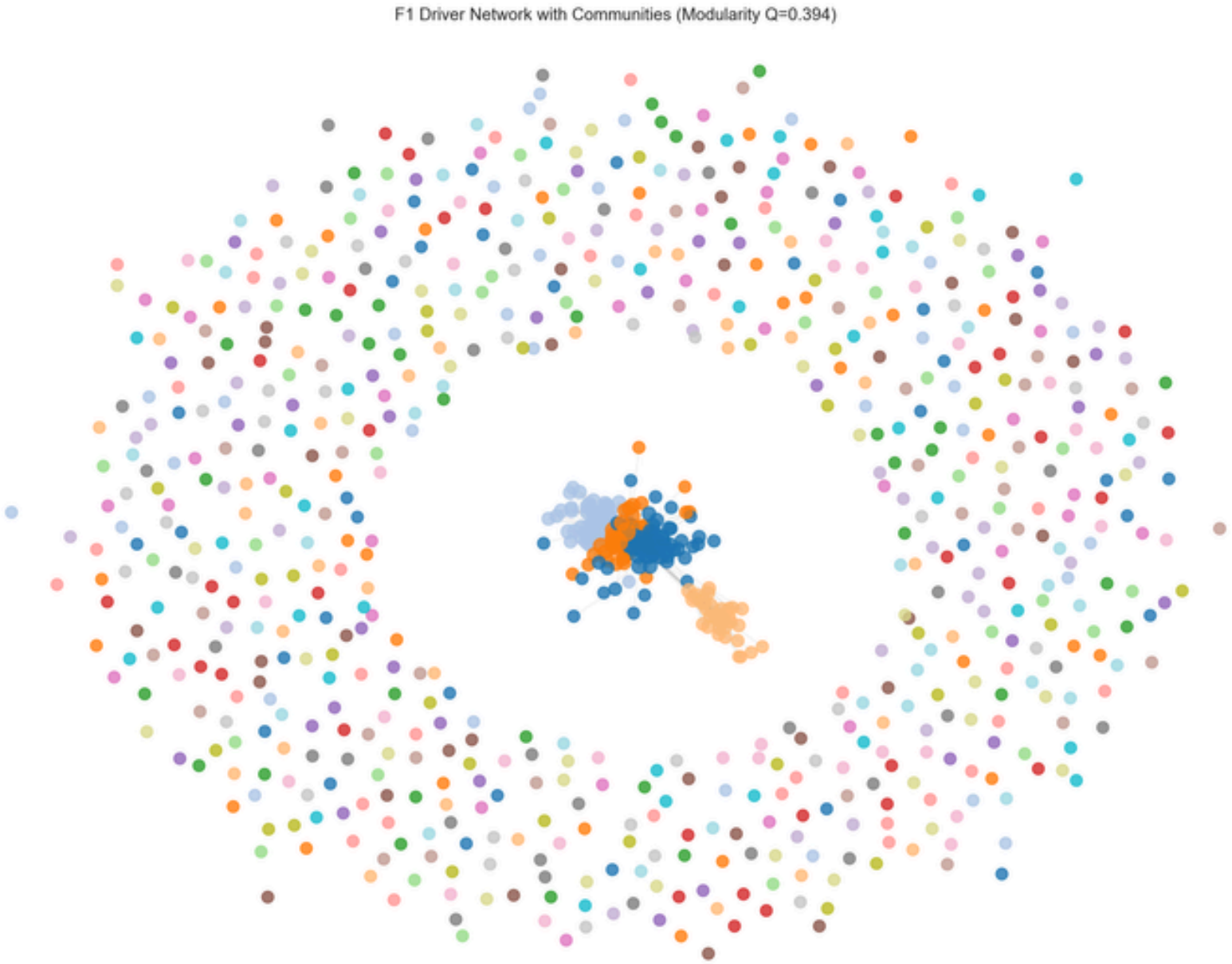
Community Detection (Louvain Algorithm)

- Applied Louvain algorithm to detect driver communities
- Detected **668 communities** with modularity $Q = 0.3945$

Community	Size	NotableDrivers
1	67	Nakajima,Sato,Speed,McNish,Brundle
2	60	Hamilton,Heidfeld,Rosberg,Alonso
3	41	Klien,Ide,delaRosa,Burti,Alesi
4	29	Grouillard,Bertaggia,McCarthy,Lammers
5–10	1each	daMatta,Badoer,Magnussen,Suzuki

Table 5: Top communities by size.

Community Network Visualization



Community Statistics

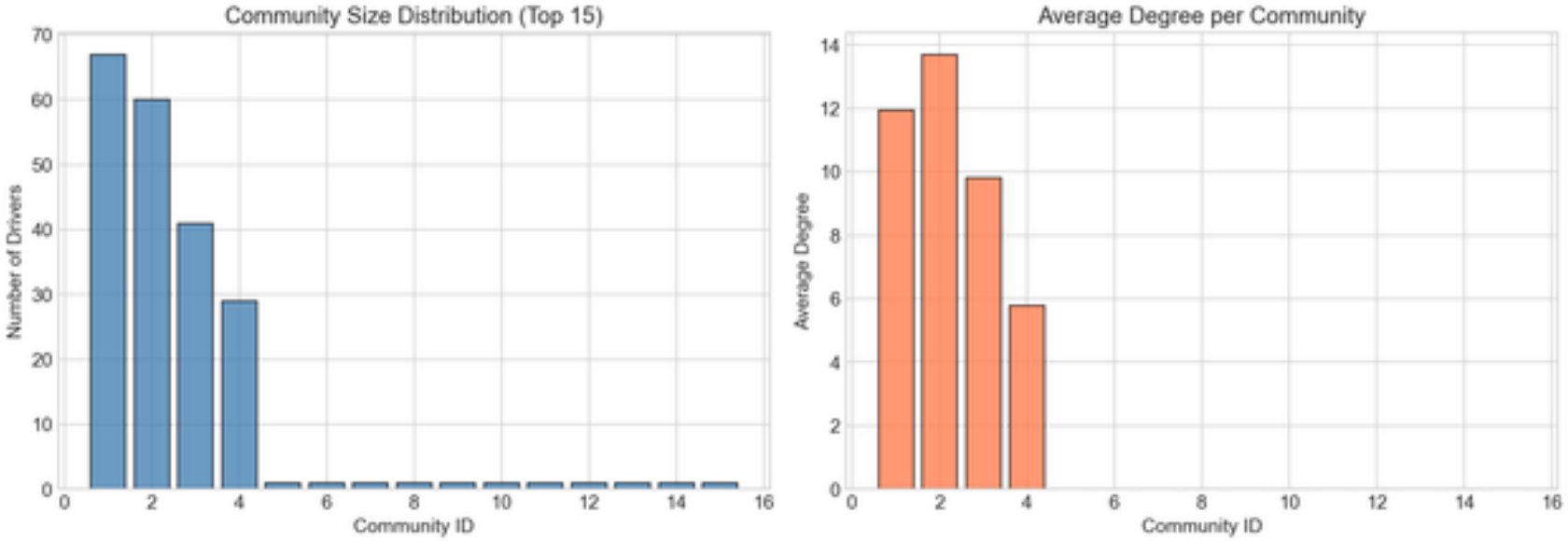


Figure 5: Community size distribution and average degree per community.

Network Robustness Simulation

- **Random Failure:** Random node removal (20 steps, 5% each)
- **Targeted Attack:** Remove highest-degree nodes first

Frac.	S_{rand}	L_{rand}	S_{targ}	L_{targ}
0.05	0.230	2.547	0.053	4.138
0.10	0.236	2.560	0.013	3.000
0.15	0.238	2.609	0.003	1.000
0.20	0.235	2.598	0.001	0.000
0.50	0.234	2.710	0.002	0.000

Table 6: Robustness results. S = Giant component size, L = Avg path length.

Robustness: Visualization

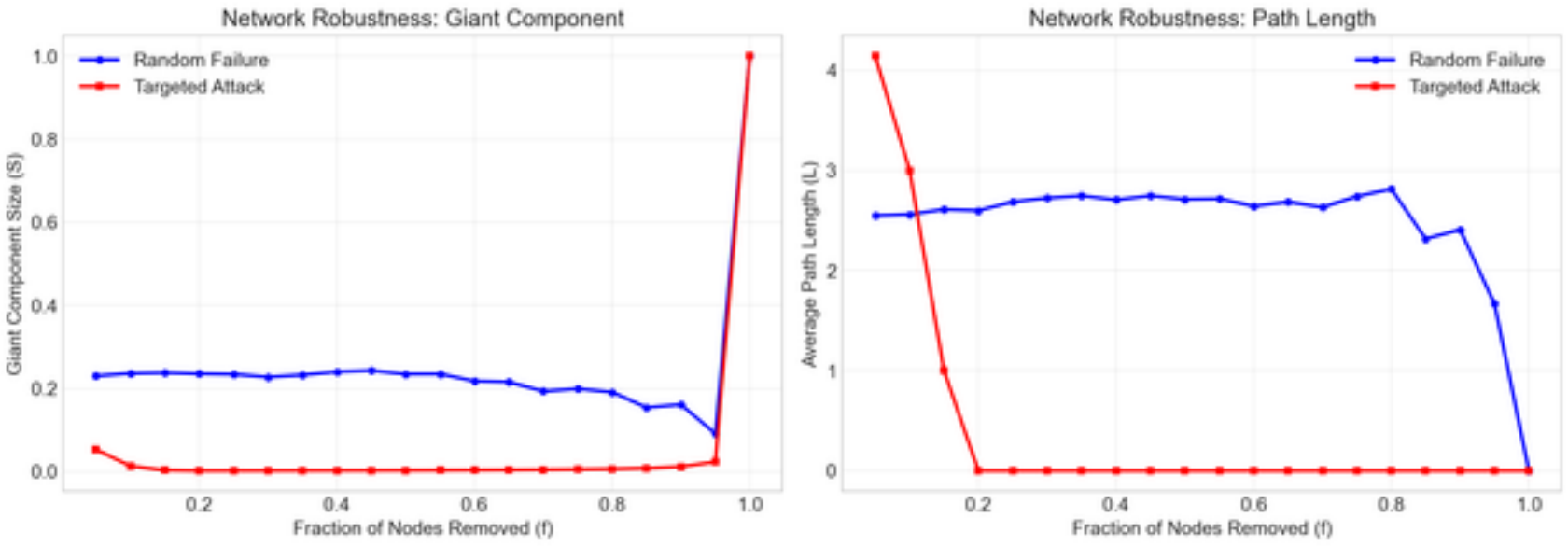


Figure 6: Giant component size and path length under random failure (blue) vs. targeted attack (red).

Technical Findings

1. **Scale-Free Property:** Power-law degree distribution ($\gamma = 0.78$)
2. **Small-World Property:** $L = 2.52 < \ln(N) = 6.76$, $C = 0.68 \gg 0.004$
3. **Hub Drivers:** Nakajima (97), Hamilton (80), Klien (74)
4. **Communities:** 668 groups ($Q = 0.3945$) reflecting F1 eras
5. **Robustness:** Resilient to random failure, vulnerable to targeted attack

Non-Technical Insights

- **Six Degrees in F1:** Any two drivers connected in ~2.5 steps
- **Reserve Driver Paradox:** Test/reserve drivers are the most connected, not champions
- **Knowledge Transfer:** Mid-tier drivers carry technical knowledge between teams
- **Vulnerability:** Removing top 5% hubs collapses network to 5.3%
- **Rich Get Richer:** Well-connected drivers attract more opp ortunities
- **Nationality Clusters:** Communities reflect national motorsport programs

Conclusion

Summary

- F1 driver network exhibits both **scale-free** and **small-world** properties
- Hub drivers (Nakajima, Hamilton) are critical for network connectivity
- 668 communities reveal distinct eras of F1 history
- Network is robust to random failures but **extremely vulnerable** to targeted attacks

Implications

Understanding network structure can inform talent development, knowledge management, and sport governance strategies in Formula 1.

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ThankYou!

Questions?